

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0795

Nunn 138 kV
April 15, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0795. The Interconnection Customer proposes a 39.9 MW natural gas combined cycle facility interconnecting at the Nunn 138 kV substation in the Pennsylvania area. The request was received on May 09, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 248 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,730,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

FOSTBURG 1-OLIVIA PARK 138kV: 137.1% loading (631 MVA) following the outage of the SPINDLE-RIDGELEY 345kV; DFAX = 9.0%.

RIDGELEY-OLIVIA PARK 138kV: 135.5% loading (634 MVA) following the outage of the SPINDLE-RIDGELEY 345kV; DFAX = 9.0%.

The FROSTBURG 1-MORBIHAN SUBSTATION 138kV is overloaded to 129.0% of its 683 MVA emergency rating for the loss of the RIDGELEY-SPINDLE 345kV. The project contributes 9.0% DFAX (3.6 MW) to this overload.

ARTESIA COUNTRY CLUB-CLARK COUNTY 345kV: 118.6% loading (1,291 MVA) following the outage of the CARL BAILEY-OTSEGO 138kV; DFAX = 7.8%.

For the loss of INTERNATIONAL PAPER VALLIANT OK-URBAN 345kV, the FRONT ROYAL-URBAN 345kV loads to 109.3% of its 1,485 MVA rating. Project DFAX: 10.2%.

The URBAN-AMERICAN 345kV is overloaded to 108.7% of its 1,903 MVA emergency rating for the loss of the RIO DELL-ST. JOHN 138kV. The project contributes 12.8% DFAX (5.1 MW) to this overload.

CLARK COUNTY-KEARNEYVILLE 345kV: 108.1% loading (1,383 MVA) following the outage of the CARL BAILEY-OTSEGO 138kV; DFAX = 8.2%.

For the loss of SPINDLE-RIDGELEY 345kV, the RIDGELEY 345 kV/138 kV Transformer loads to 105.4% of its 683 MVA rating. Project DFAX: 5.4%.

The ARTESIA COUNTRY CLUB-AIR FORCE BASE 345kV is overloaded to 104.1% of its 1,217 MVA emergency rating for the loss of the ORCHARD RD-FT HUMBUG 500kV. The project contributes 6.8% DFAX (2.7 MW) to this overload.

For the loss of RIO DELL-ST. JOHN 138kV L, the INTERNATIONAL PAPER VALLIANT OK-KEARNEYVILLE 345kV loads to 104.0% of its 1,400 MVA rating. Project DFAX: 8.1%.

For the loss of FRONT ROYAL-URBAN 345kV Line, the URBAN-INTERNATIONAL PAPER VALLIANT OK 345kV loads to 104.0% of its 1,791 MVA rating. Project DFAX: 11.6%.

For the loss of AES WARRIOR RUN COGENERATION FACILITY-RIDGELEY 138kV, the AMERICAN-RIDGELEY 345kV L loads to 102.3% of its 1,917 MVA rating. Project DFAX: 13.1%.

For the loss of FROSTBURG 1-MORBIHAN 138kV, the RIDGELEY-SPINDLE 345kV Line loads to 100.4% of its 2,006 MVA rating. Project DFAX: 13.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FROSTBURG 1-OLIVIA PARK 138kV	Rebuild line with higher-capacity conductor	\$4,821,000	2% (\$98,000)
RIDGELEY-OLIVIA PARK 138kV	Rebuild line with higher-capacity conductor	\$3,286,000	2% (\$67,000)
FROSTBURG 1-MORBIHAN SUBSTATION 138kV	Rebuild line with higher-capacity conductor	\$1,635,000	2% (\$33,000)
ARTESIA COUNTRY CLUB-CLARK COUNTY 345kV line	Rebuild line with higher-capacity conductor	\$49,632,000	0% (\$249,000)
FRONT ROYAL-URBAN 345kV	Reconductor line	\$23,508,000	1% (\$175,000)
AMERICAN SUBSTATION-URBAN 345kV	Reconductor line	\$32,726,000	1% (\$231,000)
CLARK COUNTY-KEARNEYVILLE 345kV	Reconductor line	\$21,818,000	1% (\$159,000)
RIDGELEY 345/138 kV Transformer	Replace transformer with higher-rated unit	\$14,581,000	2% (\$254,000)
ARTESIA COUNTRY CLUB-AIR FORCE BASE 345kV	Reconductor line	\$15,634,000	0% (\$63,000)
INTERNATIONAL PAPER VALLIANT OK-KEARNEYVILLE 345kV	Reconductor line	\$22,978,000	1% (\$189,000)
INTERNATIONAL PAPER VALLIANT OK-URBAN 345kV L	Reconductor line	\$9,799,000	1% (\$74,000)
AMERICAN-RIDGELEY 345kV L	Reconductor line	\$5,294,000	1% (\$39,000)
RIDGELEY-SPINDLE 345kV Line	Reconductor line	\$11,777,000	1% (\$99,000)
	Total allocated to project		\$1,730,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.